

Annual Progress Report – Jariko Project

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1. Key Progress by Product Area

1.1. Core Functionality and Interoperability

- **CALL and Parameters:** Enhanced `CALL` parameter handling to support coercion between various data types (numeric, string, DS, array). Improved `PLIST` parameter re-assignment upon program return and added support for `*INxx` as factor 1 in `PARM` for updated value reception.
- **Data Structure (DS) Operations:** Resolved issues with packed numbers in scientific notation from DB and improved `Z-ADD` for DS array fields (with/without `OCCURS`) to correctly handle offsets and casting. Fixed updating of packed numbers in DS incremented by one. Implemented `MOVEA` for array-to-array, scalar-to-array, boolean-to-array, and S-to-DS/DS-to-S assignments. Added support for `CTDATA` initialization for DS array fields.
- **/API Directive:** Addressed coexistence issues between API definitions and unnamed DS fields, and between definitions in F spec and `EXTNAME`. Added support for compile-time resolution of API variables and `LIKE`.
- **Figurative Constants and BIFs:** Implemented `%EOF` with file name, `%XFOOT`, `%LEN`, and `%DEC` for `TimeStampValue` and `DateValue`. Improved comparison of `*HIVAL` with strings and `*ZEROS` with numbers. Added support for `*START` and `*END` in `SETLL` statements.
- **Other Statements/Features:** Implemented the `EVALR` statement. Improved `COMP` between integer and `*ALL'xx'` values. Added support for `DEFINE` based on indicators. Implemented `CHECK` and `CHECKR` with expressions in factor 2.

1.2. Robustness and Stability

- **Error Handling:** Resolved `NullPointerException` during interpretation. Implemented re-throwing of `InterruptedException` even with active error indicators to ensure proper interruption. Improved error handling for `ProjectedArrayValue` and serialization to prevent `StackOverflow` issues.
- **Runtime Behavior:** Fixed issues with `MOVEL` from blank to integer. Corrected `Z-ADD` behavior when decimal/integer numbers exceed target size. Addressed `VARYING` fields in DS being longer than expected. Ensured correct initialization of DS fields without `INZ`.

- **Flow Control (GOTO , SELECT):** Fixed GOTO behavior in composite statements to prevent StackOverflow and incorrect exits. Allowed SELECT statements to contain only an OTHER branch.

1.3. Observability and Diagnostics

- **Logging:** Improved format of mock callback logs for TSV compliance. Enhanced log messages for FunctionCall with context. Added proper error messages for unimplemented TODO(...) functions.
- **Telemetry:** Introduced support for emitting Jariko and RPG telemetry traces, including profiling annotations in RPG code for tracing and analysis.
- **Callbacks:** Added onCompilationUnitEncodingError callback to handle non-blocking errors during compilation unit encoding.

1.4. Performance Optimization

- **Data Structure (DS):** Introduced the DataStructValueBuilder interface for improved performance of DS operations, yielding up to a tenfold improvement in certain tests. Optimized deserialization of packed numbers in DS.
- **Lookup Mechanisms:** Replaced list-based lookups with map-based lookups for DataDefinition , Subroutine , and FileDefinition , enhancing resolution performance.
- **Testing:** Categorized specific tests as "performance tests" for focused monitoring and analysis.
- **Substring Operations:** Improved the DataStructValue.setSubstring method, resulting in an approximate tenfold performance improvement in tests like CLEARDS01 .

1.5. Maintenance and Technological Upgrades

- **Dependency Updates:** Numerous dependency updates including log4j-core , log4j-api , org.ajoberstar.grgit , commons-io , hsqldb , com.fifesoft:rsyntaxtextarea , com.github.ziggy42:kolor , com.github.ajalt:clikt , and io.github.gradle-nexus.publish-plugin . Removed unhelpful dependencies.
- **Kotlin Upgrade:** Upgraded Kotlin version from 2.0.20 to 2.1.0 .
- **Refactoring and Code Quality:** Removed private modifiers from constructors of internal data classes to improve usability and align with Kotlin conventions. Replaced .values().first with .entries.first for enum classes. Improved exception handling and stack cleanup. Refactored test functions and ApiId loading for safer API usage.

2. Number of Pull Requests and Contributors

- **Total Number of Merged Pull Requests:** 80
- **Unique Contributors:** 8 (davidepalladino-apuliasoft, dom-apuliasoft, lanarimarco, dependabot[bot], andraasi, GaetanoDiNatale, FiorenzaBusi, paolofacchinetti)
- **Breakdown:**
 - davidepalladino-apuliasoft : 28 PRs
 - dom-apuliasoft : 32 PRs
 - lanarimarco : 10 PRs
 - dependabot[bot] : 9 PRs
 - andraasi : 2 PRs
 - GaetanoDiNatale : 3 PRs
 - FiorenzaBusi : 1 PR
 - paolofacchinetti : 2 PRs